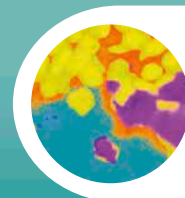
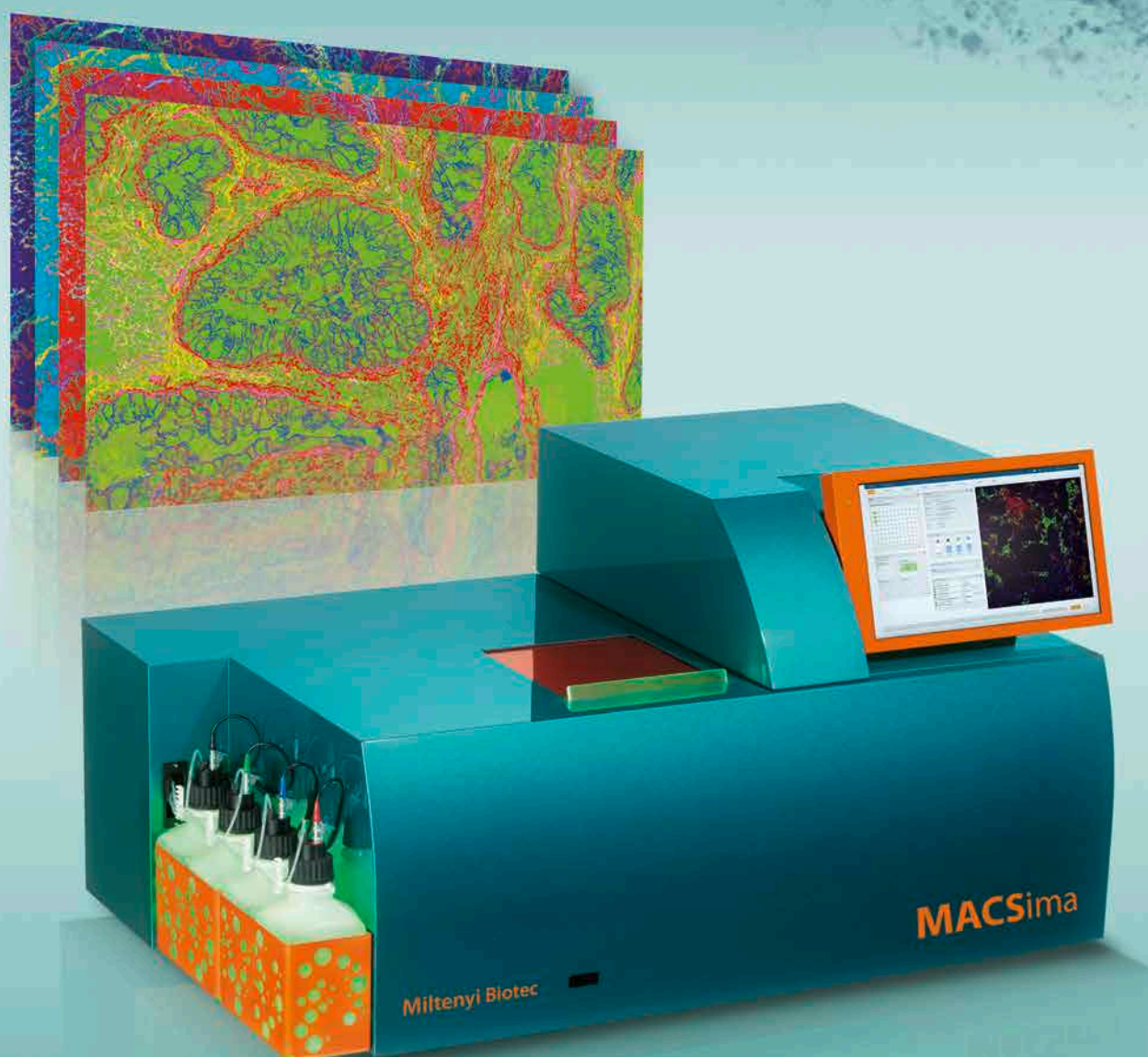




Miltényi Biotec

MACSima™ Platform

Explore the full potential of spatial biology



UNDERSTAND
NATURE'S
COMPLEXITY

The MACSima™ Platform

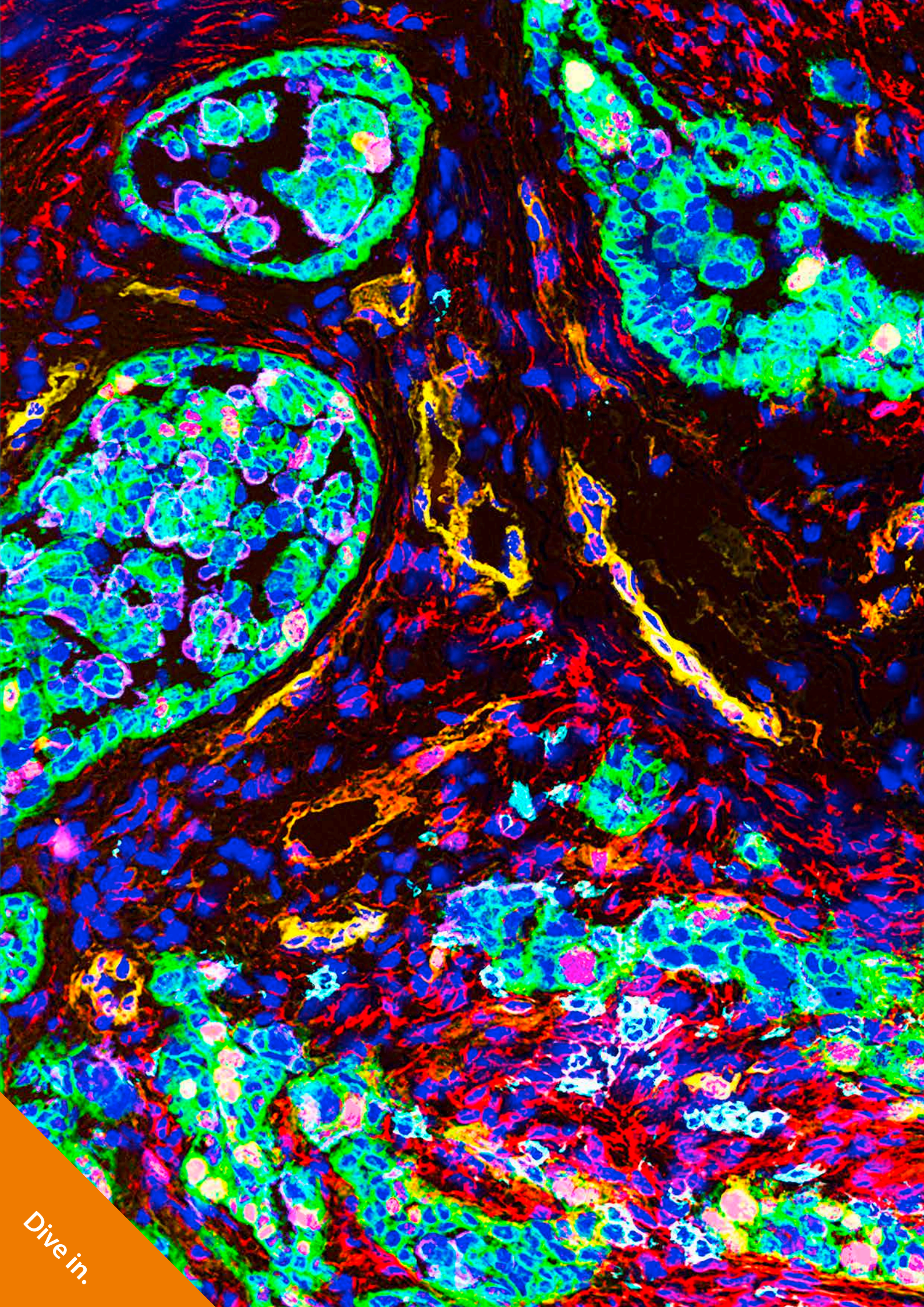
Everything you need
for fully automated,
spatial analysis

Biological systems and cellular processes are inherently complex due to the interaction of thousands of proteins involved in proper functioning of the entire organism. Thus, an in-depth analysis of biological systems requires the examination of a plethora of parameters in order to decipher the underlying principles. Currently available techniques can provide only a very limited perspective on the complexity of biological systems.

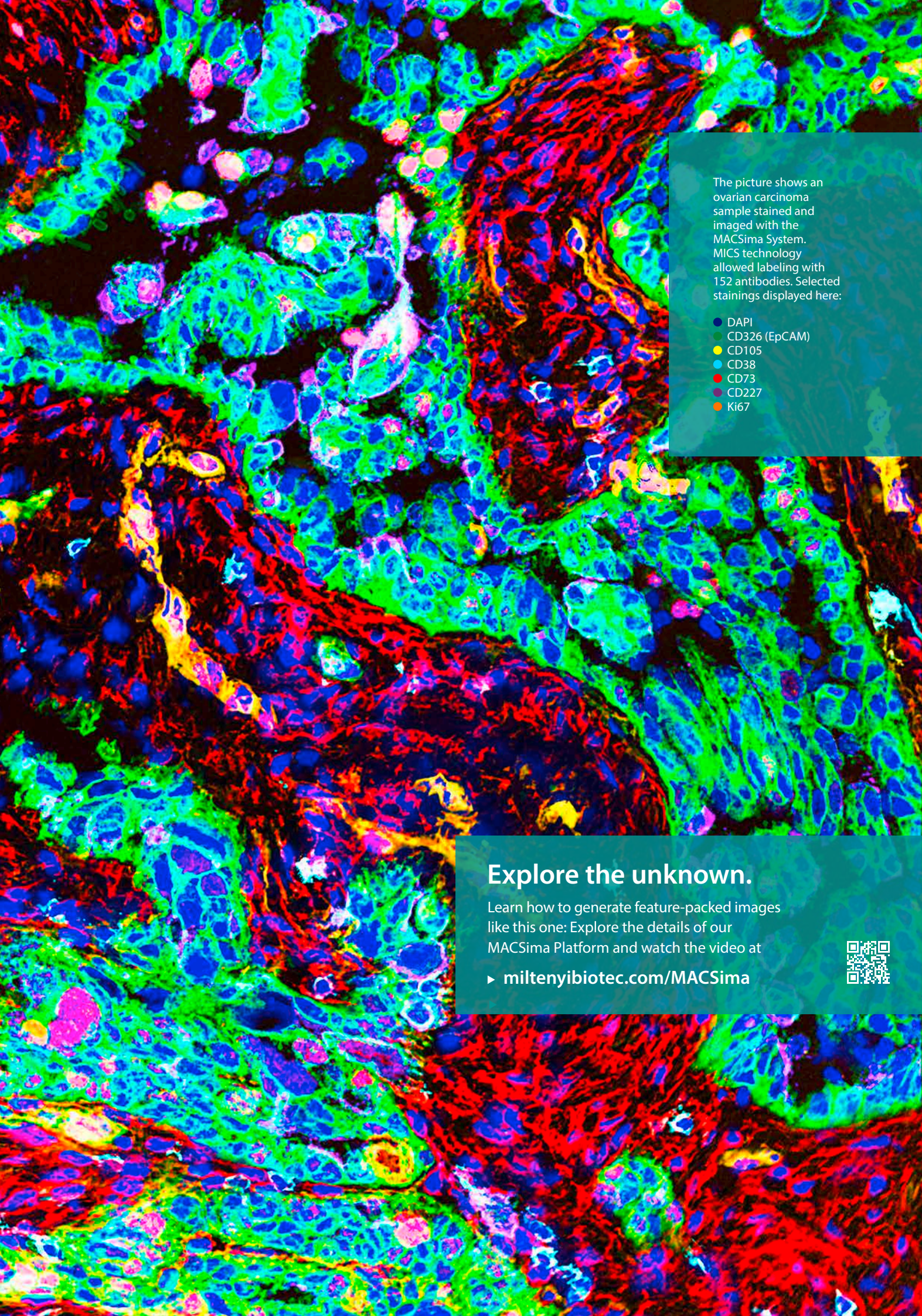
Miltenyi Biotec's MICS (MACSima Imaging Cyclic Staining) technology impressively overcomes these limits as it makes innovative use of trusted fluorescence microscopy techniques to allow the microscopic analysis of an unprecedented number of proteins or other antigens on a single sample.

Based on this technology, Miltenyi Biotec developed the MACSima Platform, which enables fully automated high-plex imaging. The possibility of evaluating localization, expression, and potential interaction of a multitude of different proteins allows scientists to tap the full potential of spatial biology.

One sample,
hundreds
of markers



Dive in.



The picture shows an ovarian carcinoma sample stained and imaged with the MACSima System. MICS technology allowed labeling with 152 antibodies. Selected stainings displayed here:

- DAPI
- CD326 (EpCAM)
- CD105
- CD38
- CD73
- CD227
- Ki67

Explore the unknown.

Learn how to generate feature-packed images like this one: Explore the details of our MACSima Platform and watch the video at

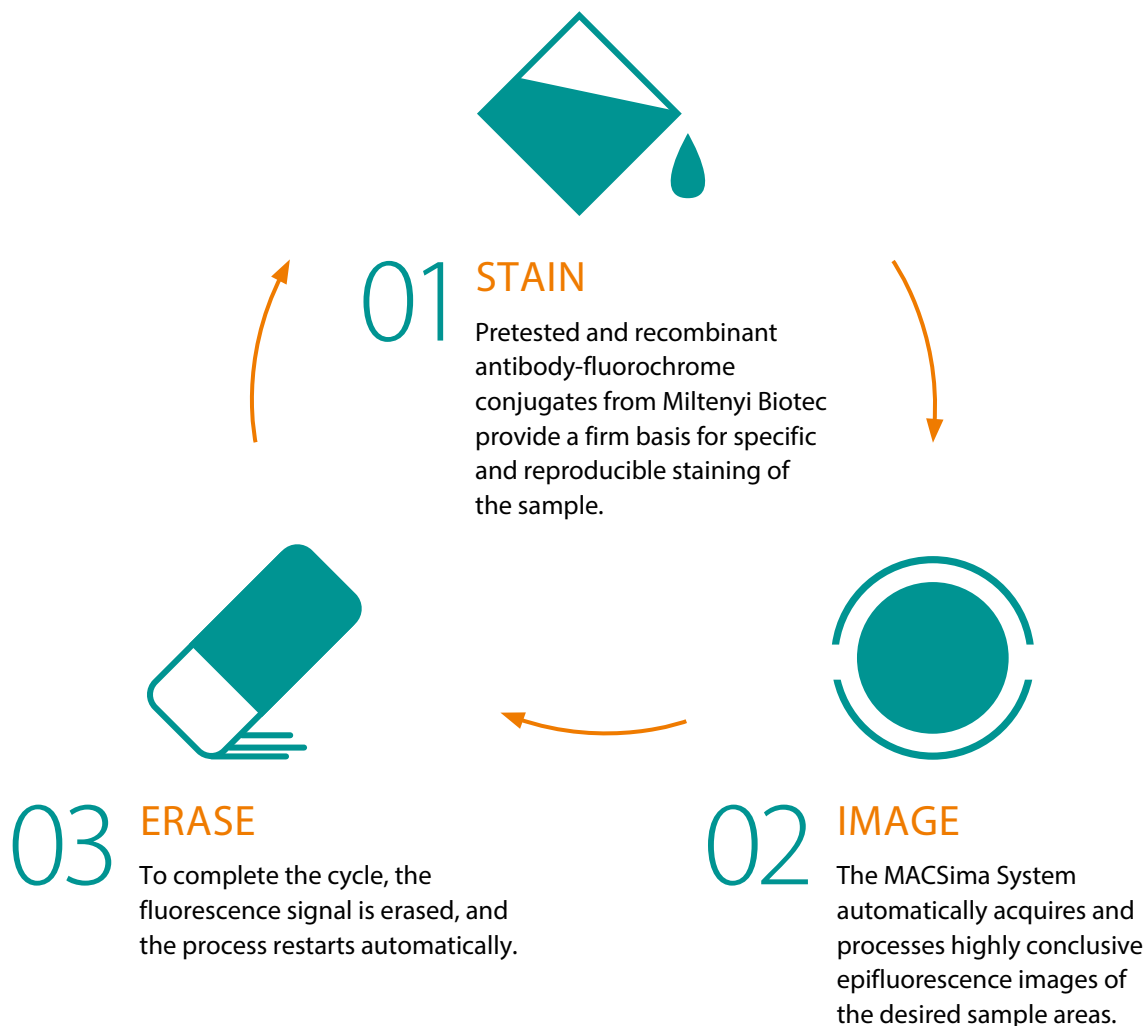
► miltenyibiotec.com/MACSima



MICS technology – the basic principle

MICS technology enables the simultaneous analysis of hundreds of markers on a single sample based on fluorescence microscopy. It uses the principle of iterative staining with different fluorochrome-conjugated antibodies to acquire microscopy data for a multitude of parameters without harming the sample. The iterative process comprises three main steps: fluorescent staining, image acquisition, and

erasure of the fluorescence signal, all of which are conducted by the MACSima System in a fully automated manner. The resulting stack of potentially hundreds of marker images provides an unprecedented insight into the physiological or pathological characteristics of the sample. Due to on-the-fly processing, data analysis can start at any time, even when the iterative process is still running.



Complete platform solution

With its automated processes and optimized components, the MACSima Platform reduces the effort required for the generation of complex spatial biology data to the basic essentials. The benefits of this complete, harmonized solution speak for themselves.

Hundreds of markers on one sample

Obtain high-plex spatial data to analyze hundreds of proteins and other antigens on a single sample.

Fully automated instrument

Plan your experiment and leave the execution to the MACSima System.

Broad portfolio of pretested antibodies

Analyze hundreds of markers on your sample based on a plethora of recombinantly engineered, fluorochrome-conjugated antibodies, tested specifically for MICS.

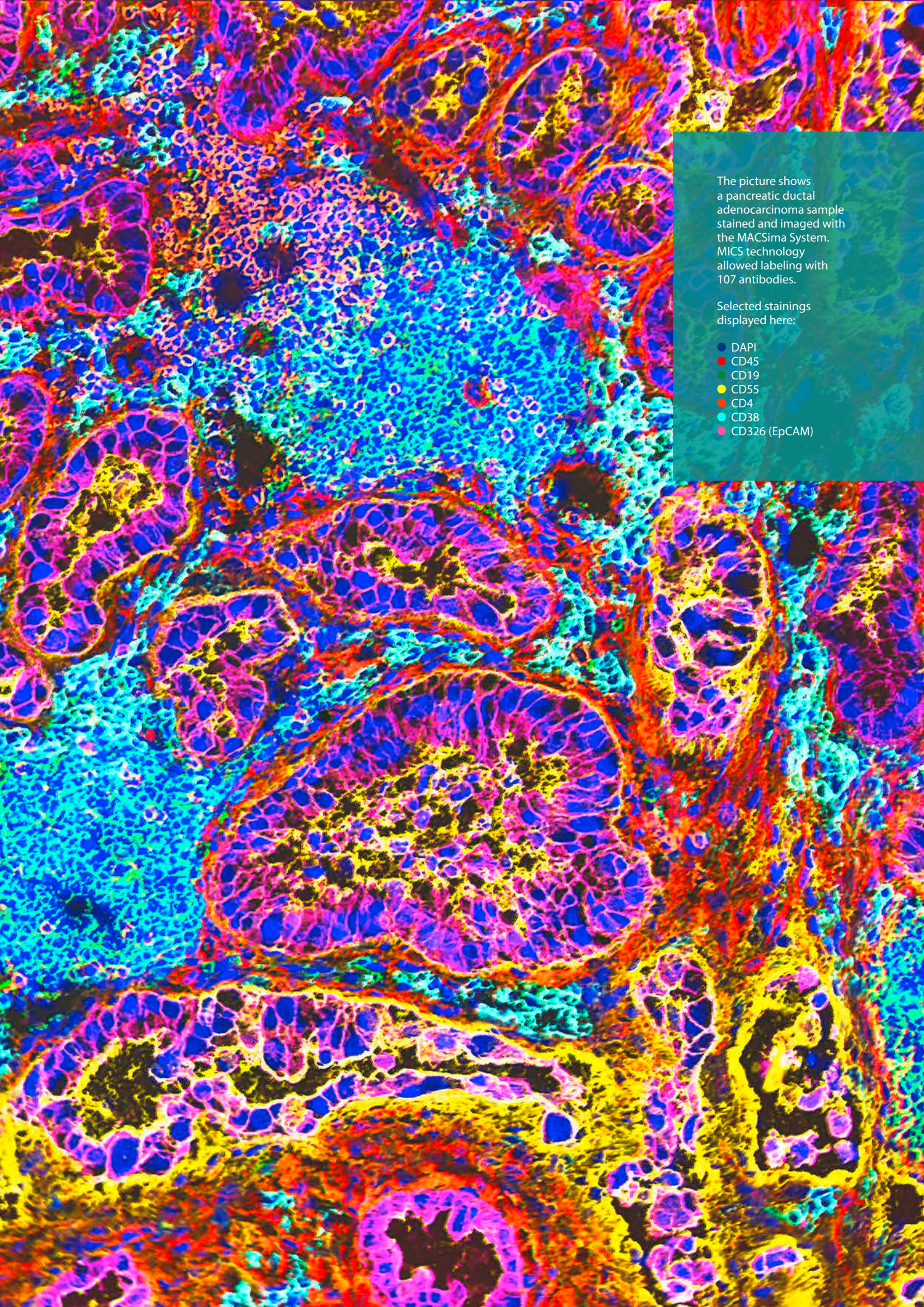
Effortless experiment preparation

Benefit from plug-and-play REAscreen™ Antibody Panels containing pre-defined sets of antibodies from Miltenyi Biotec for an effortless yet comprehensive analysis. Analyze any kind of fixed sample with our MACSwell™ Sample Carriers and benefit from various formats of antibodies for full flexibility.

Sophisticated yet simple analysis software

Analyze your high-plex spatial data easily and comprehensively using MACS® iQ View. Its versatility and intuitive user interface make this image analysis software the perfect companion for your spatial biology experiments.





The picture shows a pancreatic ductal adenocarcinoma sample stained and imaged with the MACSima System. MICS technology allowed labeling with 107 antibodies.

Selected stainings displayed here:

- DAPI
- CD45
- CD19
- CD55
- CD4
- CD38
- CD326 (EpCAM)

Spatial biology made easy



1 Set up experiment

Prepare your sample and antibody panel, taking your preferred fixation method into account.



2

Start experiment

Place your sample and reagent plates in the MACSima System and define your preferred imaging area. Start the experiment and leave the iterative MICS technology-based process to the MACSima System.



3

On-the-fly data analysis

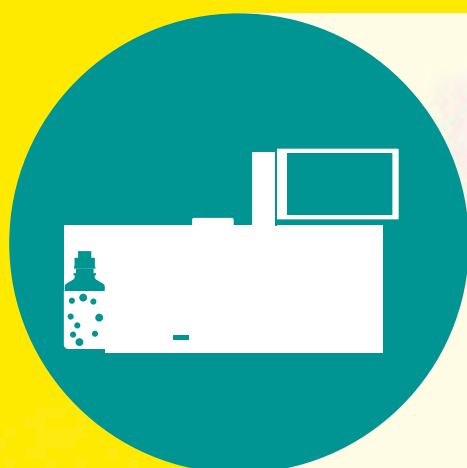
Start analyzing your acquired data, even if the experiment is still running.

MACSima

Integrated system allows for effortless and reliable processing

MICS technology is based on the well-established and straightforward technique of immunofluorescence staining. However, to translate this technique into comprehensive and effortless multiparameter imaging,

a functional, well-orchestrated system is a prerequisite. The MACSima Platform relies on four essential pillars that are combined seamlessly to ensure the easy generation of conclusive spatial data.



MACSima SYSTEM

At the heart of the platform is the MACSima System, an instrument that performs the iterative staining cycle and data processing in a fully automated fashion.



PRETESTED ANTIBODIES FOR MICS

An extensive and continuously growing portfolio of pretested antibodies for MICS, including recombinantly engineered antibodies, ensures specific staining and reliable analysis of hundreds of markers.

MAC Platform

Sima

MACSwell™ SAMPLE CARRIERS

Specially designed sample carriers from Miltenyi Biotec ensure a failure-free, automated MICS process and offer the flexibility to use various sample types.



MACS® iQ VIEW ANALYSIS SOFTWARE

Specifically developed to analyze the enormous data stacks obtained by the MACSima System, this software is the perfect tool to dissect the multidimensional datasets and to extract the underlying information in an easy and comprehensive manner.



MACSima System

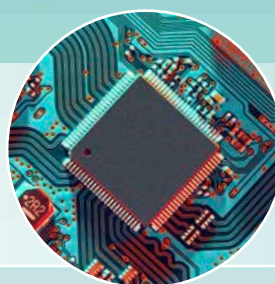
Fully automated sample handling and data acquisition

Excellent optics, a state-of-the-art sCMOS camera, accurate liquid handling, and high computer capacity are the hallmarks of the MACSima System.

All system components are perfectly geared to each other, allowing truly automated and conclusive spatial analysis based on MICS technology.

Computers

Two computers with high storage capacity allow for detailed data analysis while the image stack continues to grow cycle after cycle.



Liquid handling system

The robotic needle arm eliminates pipetting errors and saves valuable time as it can handle hundreds of antibodies and all other required liquids accurately and fully automatically. Automated washing processes prevent carryover of reagents to maintain fluorescence signal specificity across all images.



Ultraprecise stage

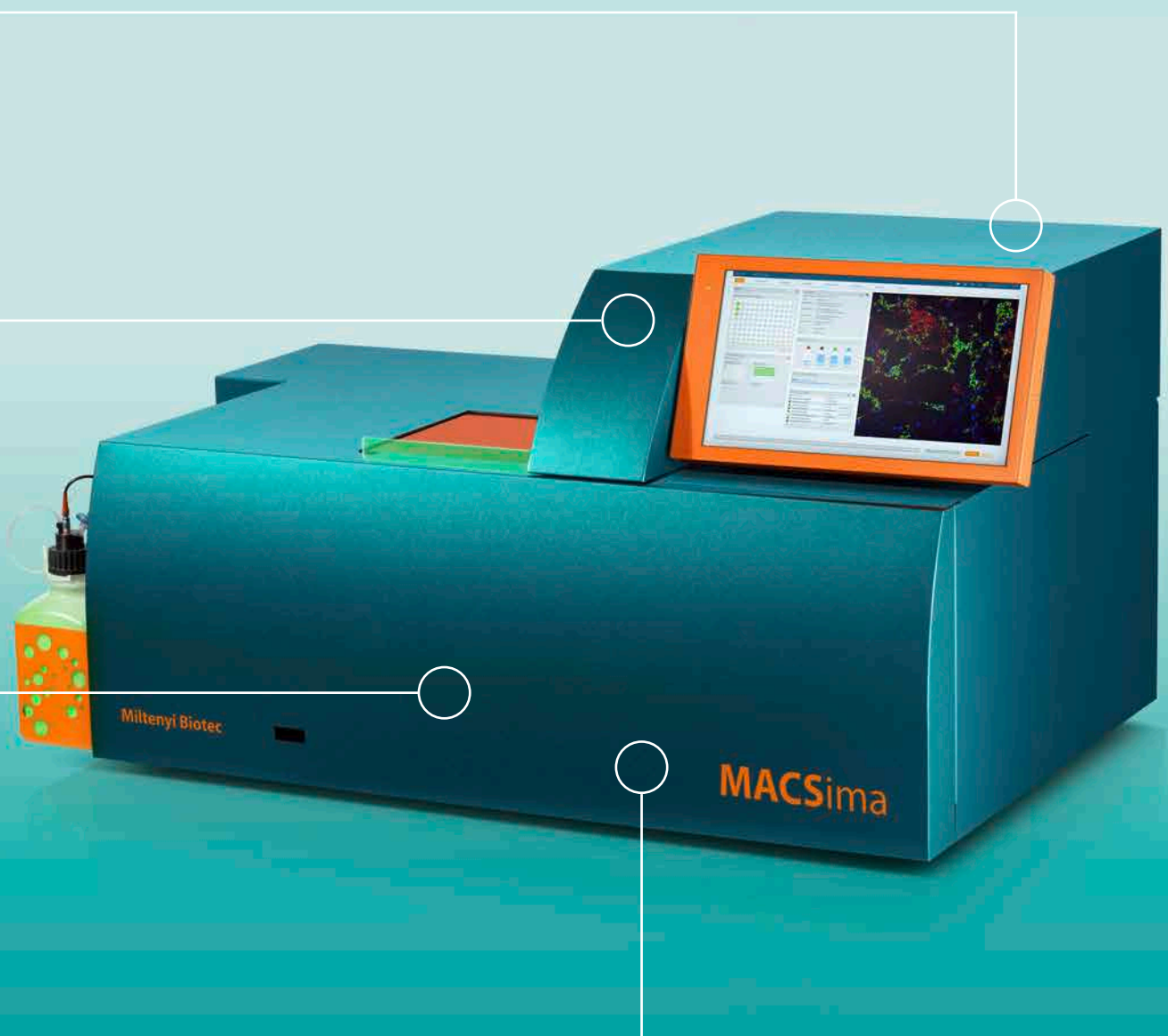
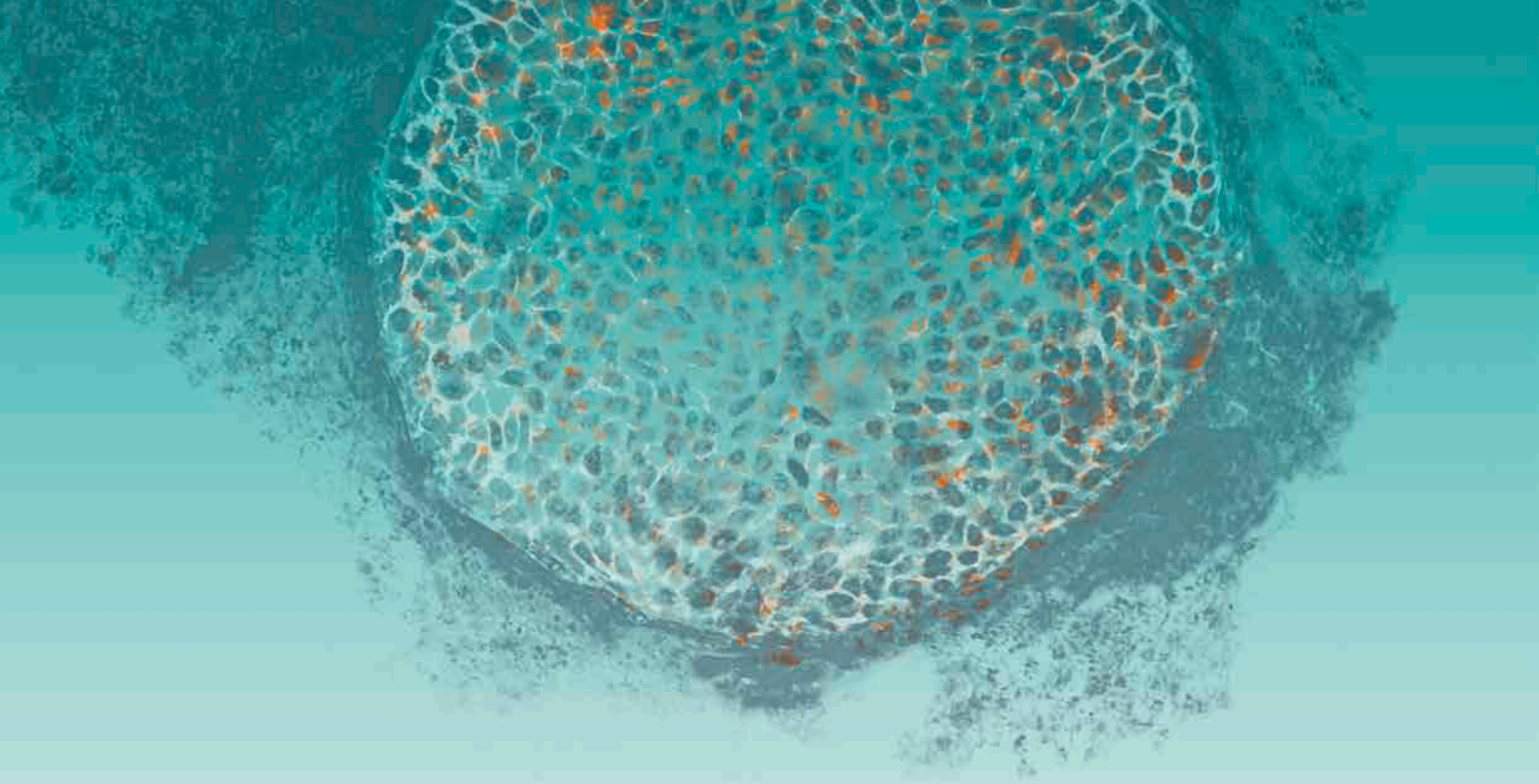
Hosting the sample and all reagents, the ultraprecise stage serves as the MACSima System's work bench. Accurate positioning of all components throughout the MICS experiment ensures reliable execution of the iterative process and exact maintenance of the field of view.



State-of-the-art microscope and camera

The widefield microscope includes one objective for a fast overview scan. Two 20x objectives enable optimal imaging of subcellular details, regardless of the sample carrier used – be it long working distance with microscope slides and MACSwell Imaging Frames or short distance with MACSwell 24 Imaging Plates. The sCMOS camera provides excellent image quality.





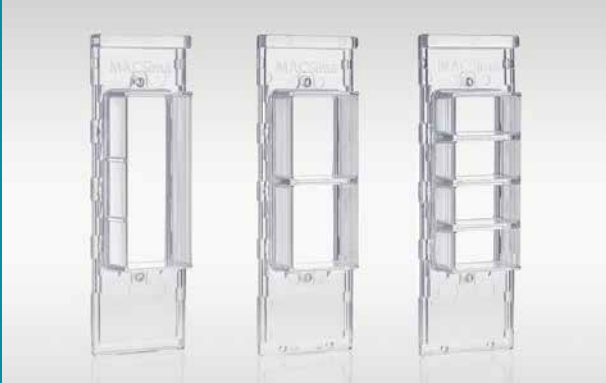
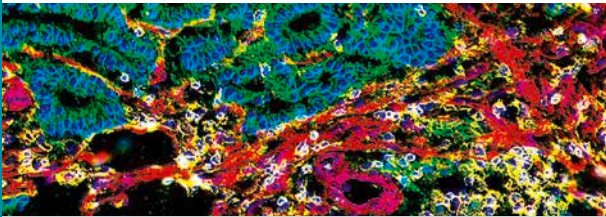
MACSwell™ Sample Carriers

Analyze any kind of fixed sample

To answer complex scientific questions you can't afford to be restricted by technical limitations. To give you complete flexibility in the type of fixed sample you want to analyze with the MACSima System, we have developed the MACSwell

Sample Carriers. To support either tissue, adherent or suspension cells, we have designed various types of devices. Each of them contains a well-defined reaction cavity to perform MICS experiments easily and safely and assure you failure-free experimentation.

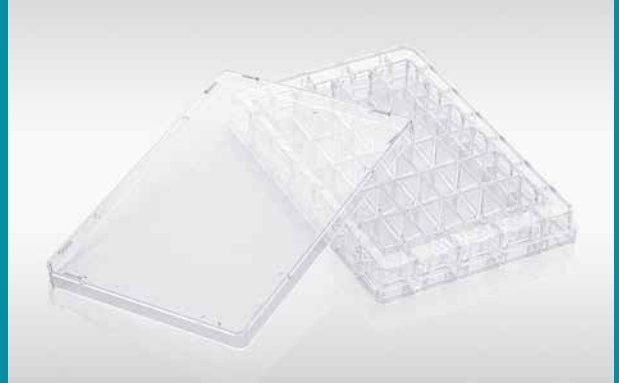
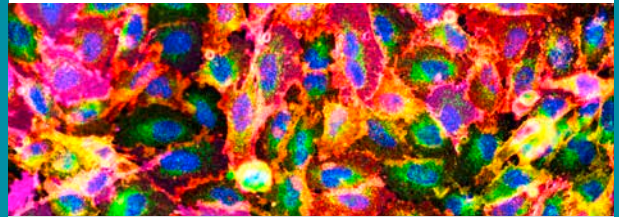
TISSUE



MACSwell Imaging Frames

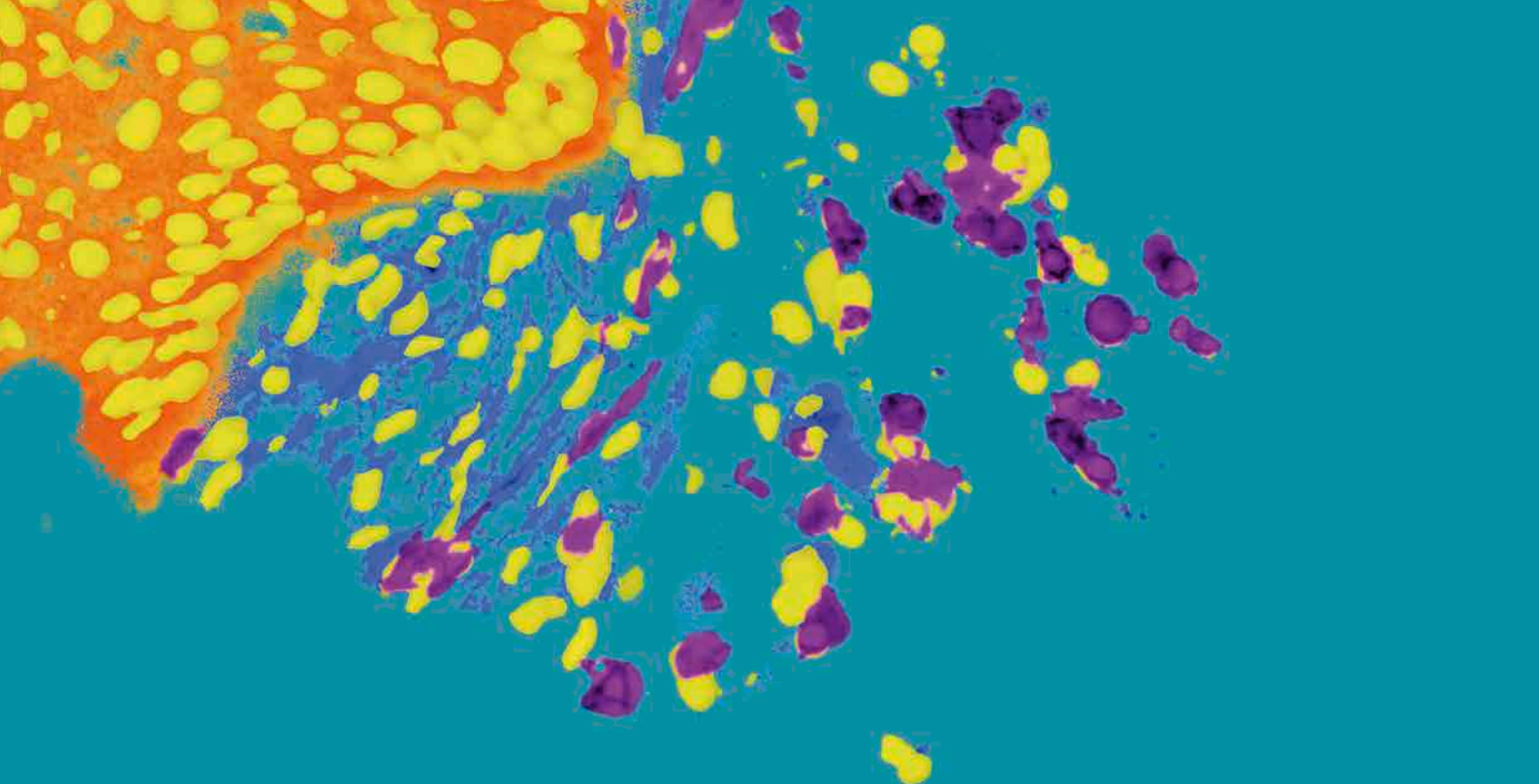
can be mounted around any standard microscope slide and provide the reaction cavity needed for a MICS experiment. MACSwell Imaging Frames are provided with various different sizes of reaction cavities to perfectly fit the size of your tissue sample.

ADHERENT CELLS

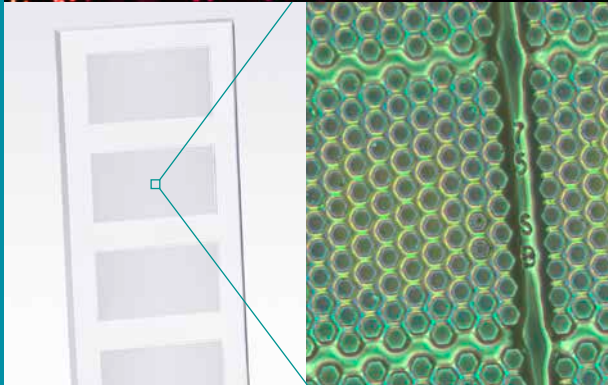
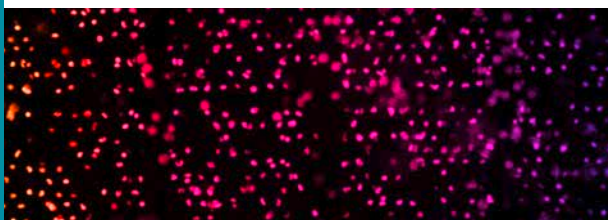


MACSwell 24 Imaging Plates

are pre-assembled and contain 24 rectangular wells with a clear 170 µm thick cover glass bottom. Simply pipet your cell suspension onto the plate, culture as usual, and fix the sample according to your standard protocol directly on the plate.



SUSPENSION CELLS

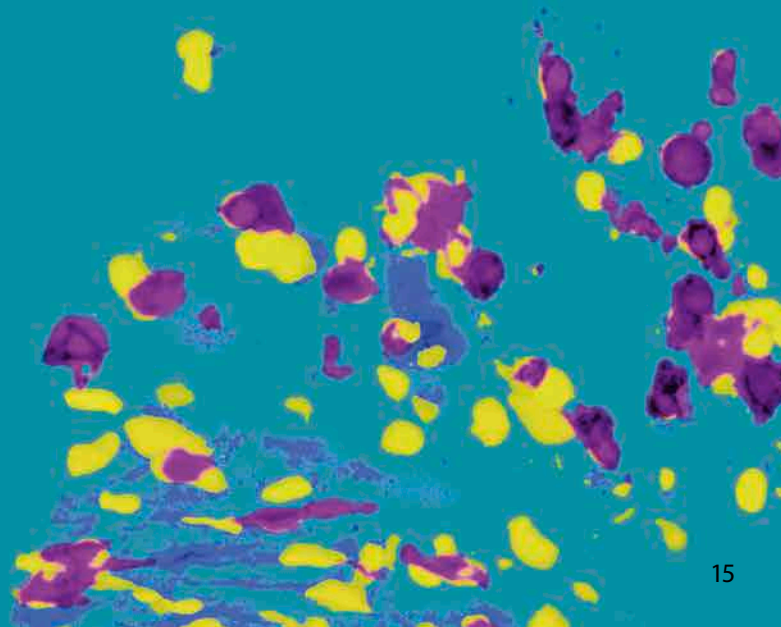


MACSwell Sample Carriers with microcavities

MACSwell Micro Slides contain over 1.5 million hexagonal microcavities, which are perfectly sized to fit exactly one cell. Simply mount your choice of MACSwell Imaging Frame around the slide and pipet your suspension cells in the resulting reaction cavity. For even greater flexibility, MACSwell 24 Micro Imaging Plates complete the portfolio of sample carriers featuring microcavities.

VALIDATED FIXATION METHODS

- ✓ Formalin-fixed paraffin-embedded (FFPE)
- ✓ Paraformaldehyde (PFA)
- ✓ Acetone



Pretested antibodies for MICS

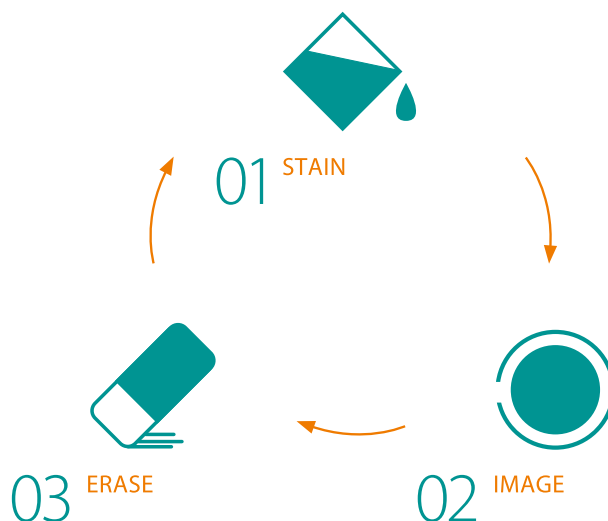
Specific staining – conclusive results

If you want to achieve high reproducibility and error-free analysis, rely on recombinant antibodies.

- World's largest portfolio of antibodies for spatial analysis
- Recombinant antibodies specifically tested for MICS technology
- Tested for compatibility with FFPE-, PFA-, or acetone-fixed samples of human or mouse origin
- Lot-consistent and reproducible results due to sophisticated recombinant antibody technologies

The two mechanisms for signal erasure

After staining with fluorochrome-conjugated antibodies (01) and image acquisition of the stained sample (02), the fluorescent signal can be erased by either of the two mechanisms shown below.



The fluorescence signal of samples that were stained with fluorochrome-conjugated antibodies, such as our recombinant REAfinity™ Antibodies coupled to non-photostable fluorochromes, can be erased via photobleaching.



Staining of samples with REAdye_lease™ and REAlease® Fluorochrome-Conjugated Antibody complexes allows for fast and gentle signal erasure via a controlled release of fluorochromes.



Convenient antibody formats



Antibody conjugates for MICS

Choose from a huge portfolio of MICS-pretested antibodies and a variety of fluorochromes to design your panel flexibly.

Explore 
500+
antibodies

► miltenyibiotec.com/MICS-antibodies



REAscreen™ Standardized Antibody Panels

Plug-and-play panels of dried MICS-pretested antibody conjugates are designed for maximum convenience and reproducibility. Whether with panels developed by Miltenyi Biotec for specific applications or with customer-designed panels (REAscreen Design), these plates save you time and effort as they eliminate the need for tedious and error-prone manual pipetting.

MACS® iQ View

Analysis software that is sophisticated yet simple



MACS iQ is a family of easy-to-use, powerful software solutions that unlock the full capabilities of Miltenyi Biotec instruments. As the first family member, MACS iQ View was specifically developed to analyze the unprecedented amount of data created by the MACSima Platform. MACS iQ View makes handling and analysis of hundreds of images as easy as it gets: Conveniently display all experimental markers individually or in any combination you need. Organize your data efficiently based on a variety of display options. Save your preferences to increase the consistency and reproducibility of your analyses.

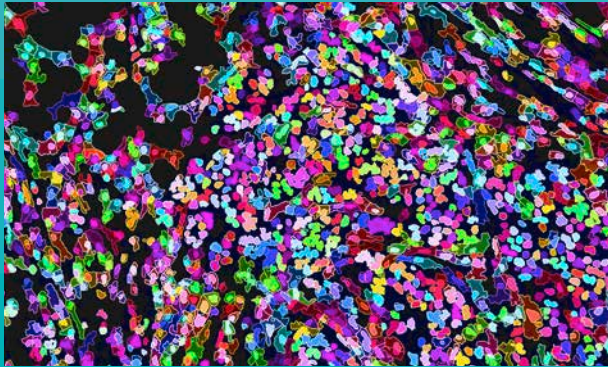
Interactive and dynamic data display

The software's user interface is highly intuitive and easy to use. Interactive and dynamic gating or clustering enables you to view the results immediately in a variety of graphs, tables, and a vast array of plots.

Visit our webpage to learn more about the MACS iQ View Software.

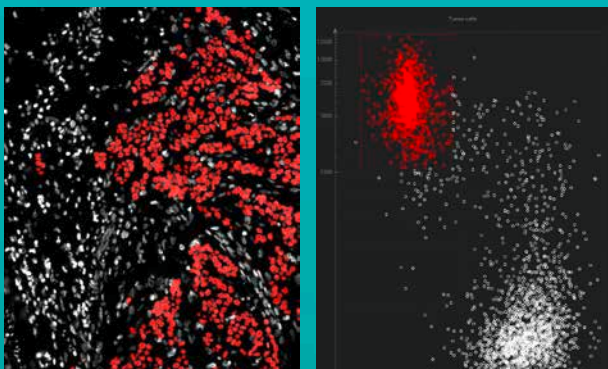
► miltenyibiotec.com/MACS-iQ-View





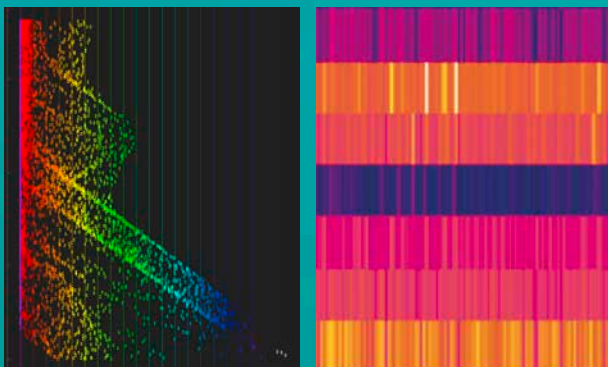
Easy segmentation

Defining the cell and nucleus boundaries is an easy task with the MACS iQ View Software. Simple definition of parameters and fast processing let you start with your analysis at once. Choose between various pre-defined segmentation options or conveniently import your own trusted segmentation mask.



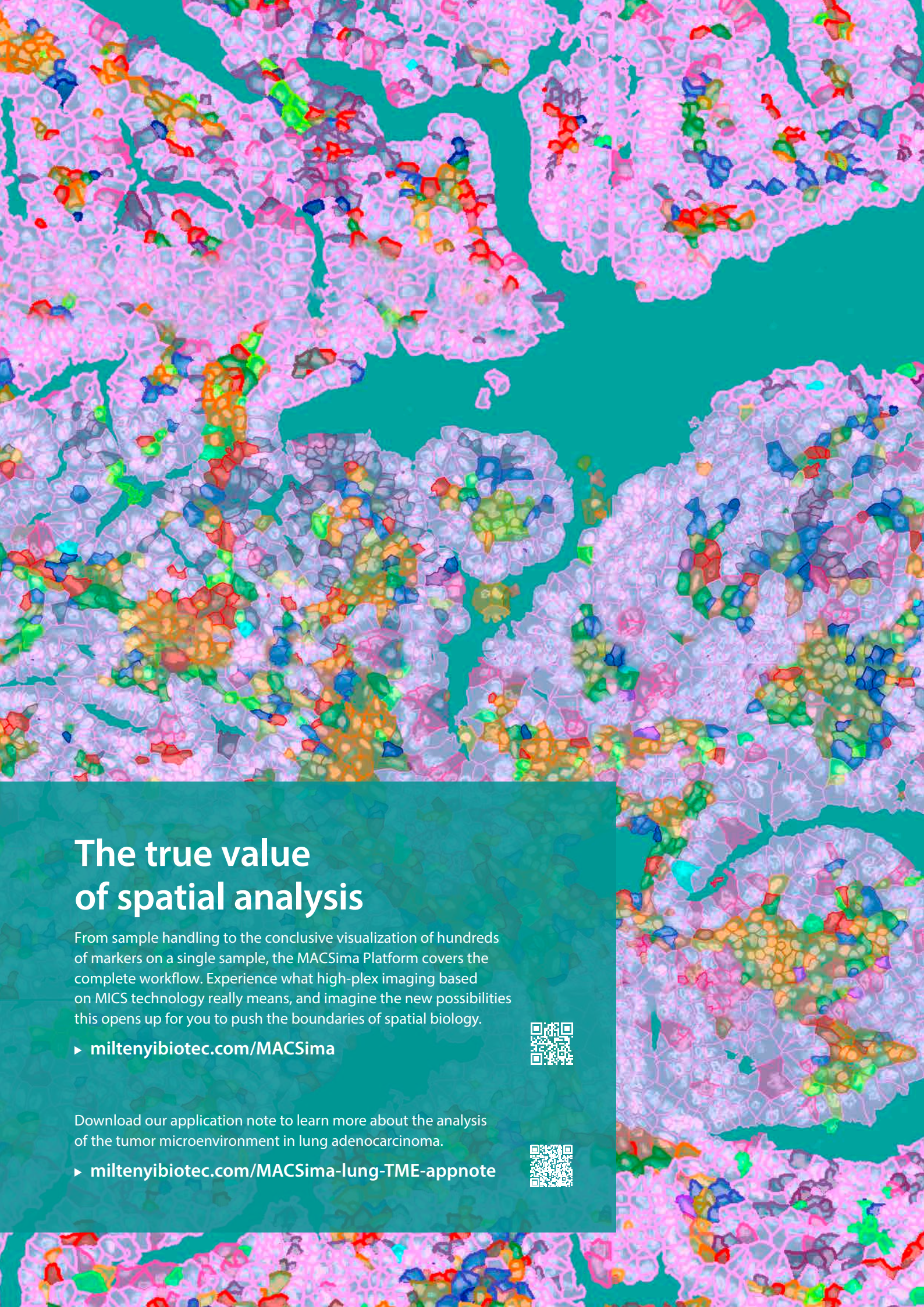
Multifaceted gating and clustering enable deep phenotyping

Deep phenotyping can be achieved through highly interactive, dynamic manual gating both on the graph and the image. A workflow editor lets you keep track of all selection steps. Alternatively, you can use the integrated clustering options according to the features of your interest. All this allows you not only to phenotype the cells but also view their position on the image and analyze spatial relationships and more.



Cutting-edge analysis and plotting options for clear data display

MICS data enable you to examine tissue samples and cells in great detail. To display the high-dimensional data on the cells' complex phenotypes and spatial organization in an easy-to-grasp way, the software offers sophisticated analysis tools. Dimension reduction plots (t-SNE and UMAP), distance mapping, and a range of additional display options such as heatmaps let you easily illustrate your intricate data.



The true value of spatial analysis

From sample handling to the conclusive visualization of hundreds of markers on a single sample, the MACSima Platform covers the complete workflow. Experience what high-plex imaging based on MICS technology really means, and imagine the new possibilities this opens up for you to push the boundaries of spatial biology.

► miltenyibiotec.com/MACSima



Download our application note to learn more about the analysis of the tumor microenvironment in lung adenocarcinoma.

► miltenyibiotec.com/MACSima-lung-TME-appnote





The picture shows a sample of lung adenocarcinoma that was stained with more than 125 antibodies to characterize the tumor microenvironment in detail. Identification of more than 20 known cell populations based on more than 40 established markers enabled the dissection of intricate marker combinations for a deep phenotyping of the sample. However, the option to use such a large number of antibodies on a single sample also allows the discovery of as yet unknown cell populations.

- CD4⁺ T cells
- CD8⁺ T cells
- Dendritic cells
- NK cells
- Monocytes
- Myeloid cells
- Neutrophils
- Granulocytes
- Macrophages
- M1 macrophages
- M2 macrophages
- Cancer stem cells
- Tumor cells
- Blood vessels
- Antigen-presenting cells
- Anti-apoptotic cells
- Activated T cells
- Plasma cells
- B cells
- Proliferating cells
- Fibroblasts
- Activated fibroblasts

One experiment, myriads of findings



Deep phenotyping

Characterization of hundreds of markers on all cells of a tissue sample enables you to identify a multitude of different cell types and explore their spatial relationships. These data provide you with an immense resource for deep phenotyping studies.



Biomarker research and discovery

Understanding how the different markers correlate in your sample is crucial for the discovery of biomarker signatures. Ultimately, these findings can help you to develop patient stratification strategies and novel therapies for many diseases.



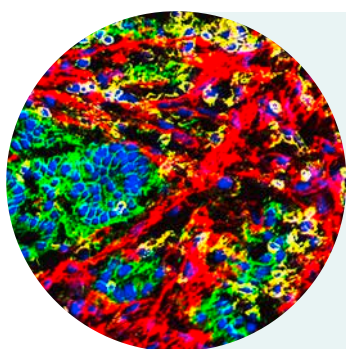
Drug target discovery and validation

Ultrahigh-plex analysis of this huge number of markers also allows you to identify potential drug targets. Resulting candidates can then be further analyzed for their effectiveness and validated in the spatial context of the full microenvironment in preclinical research studies using the MACSima Platform.

High-plex imaging in action

The MACSima Platform revolutionizes the world of high-plex imaging. Comprehensive analysis of a vast number of markers not only allows you to enhance your present research; the data will also provide you with a tremendous archive to answer future questions that you haven't even thought

about at the time of your experiment. You can go back to the complete datasets from your MICS experiment at any time and look at them from a new perspective. The sky is the limit when it comes to ultrahigh-plex imaging in spatial biology. Here are some examples.

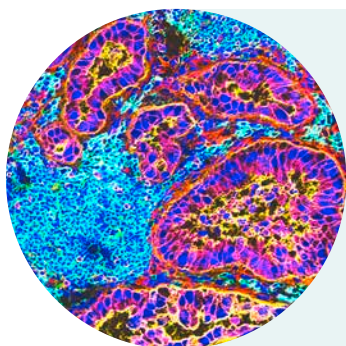


Spatial single-cell atlas of the liver

MICS technology helped researchers to generate a cellular atlas of the entire human and murine liver.

Read the paper in Cell.

► [cell.com/cell/fulltext/S0092-8674\(21\)01481-1](https://cell.com/cell/fulltext/S0092-8674(21)01481-1)

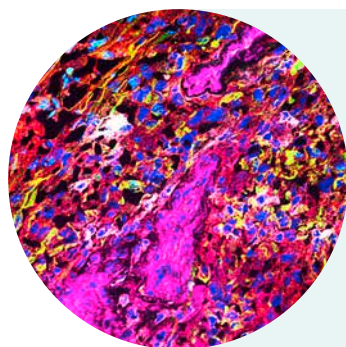


Identification of target candidates for CAR T cell-based immunotherapy of pancreatic adenocarcinoma

Pancreatic ductal adenocarcinoma samples were fixed with acetone and screened based on 107 markers.

Read the publication in Nature Communications.

► nature.com/articles/s41467-021-21774-4



New potential target pair for CAR T cell treatment of solid tumors

In this publication, more than 300 antibodies were applied to analyze a single PFA-fixed tissue sample.

Read the full story.

► nature.com/articles/s41598-022-05841-4



Take a look at our scientific content featuring the MACSima Platform.

► miltenyibiotec.com/MACSima-resources



All you need for your spatial biology experiment

MACSima System specifications

Microscope

Instrument type	Widefield epifluorescence microscope
Camera	Equipped with a latest-generation sCMOS camera with high resolution and thermoelectrically cooled sensor (15 megapixel, large 25 mm diagonal field of view)
Excitation	Six high-power excitation LEDs (filters: 386/23 nm, 420/10 nm, 470/40 nm, 531/46 nm, 628/32 nm, 725/40 nm)
Emission	Five emission filters (470/40 nm, 530/43 nm, 580/25 nm, 698/70 nm, 809/81 nm)
Autofocus	Dual approach of hardware- and image-based autofocus mechanisms
Objectives	• 2× objective to generate overview images; NA 0.1 • 20× long working distance objective (designed for 1 mm thick slides); NA 0.45 • 20× objective with high numerical aperture (designed for 170 µm thick cover glass); NA 0.75
Bleaching unit	• Separate bleaching unit for optimized signal erasure • Illuminated area: 3 mm × 3 mm • Light intensity: 2 W

Liquid handling system

Needle arm	Robotic needle arm allowing fully automated liquid transfer
Steel needle	Washable stainless steel needle with minimized spillover
Syringe pump	Syringe pump drive for accurate volumetric staining of samples
Fluid containers	1.5 L fluid containers for automatic operation over several days without the need to exchange containers

Sample and reagent stage

Automated stage	Stage with sub-micron positioning accuracy of 100 nm (automated image registration for different cycles)
Automated sample and reagent carrier positioning	Convenient loading and clamping of all supported sample carriers and reagents

Technical data

Computer	Two integrated computers for experiment planning/execution and image analysis
Monitors	• Tilttable integrated touch display (Full HD) for PC1 • External 4K monitor for PC2
Storage/ports	• 25 TB integrated storage to store data of several experiments • 2× USB 3.0 • HDMI port
Network	1× 10 GbE 1× RJ45 GbE/10GbE
Power requirement and consumption	100–240 V, 50/60 Hz, max. 7/13A, max. 1,300 W

MACS iQ View Software

Control software	• Comprehensive sample and reagent management system • Easy planning and design of both simple and complex experiments • User-friendly execution and monitoring of experiments
Analysis software	• Easy display of hundreds of markers from high-dimensional datasets • Fast and flexible segmentation • Interactive gating • Unbiased data analysis (k-means clustering, UMAP and t-SNE calculation) • Multiple plotting options (histogram, scatter plot, strip and violin plots, heatmaps) • Workflow editor • Distance analyses

Size and weight

Dimensions	• Instrument dimension: 1,220 mm × 780 mm × 650 mm (w × d × h); footprint: 1,210 mm × 750 mm (w × d) • MACSima Table¹: 1,250 mm × 765 mm × 813 mm (w × d × h)
Weight	• Instrument: 170 kg • MACSima Table¹: 120 kg

Product type	Order no.
Instrument	
MACSima System	130-121-164
MACSima Table	150-001-781
MACSima 4K monitor	150-001-775
Software	
MACS iQ View Analysis Software Licenses ²	
• Annual license	
• Permanent license	
Antibodies and reagents	
MICS-pretested antibody conjugates ³	
• REAfinity Recombinant Antibodies	see website
• REA dye_lease Releasable Fluorochromes	
• REAlease Releasable Antibodies	
Pre-defined Antibody Panels for MICS ³	
• REAscreen MAX Kit, human, FFPE	see website
• REAscreen MAX Kit, human, PFA	
• REAscreen MAX Kit, mouse, PFA	
Support Reagents for MICS	
• MACSima Stain Support Kit, human	130-127-574
• MACSima Stain Support Kit, mouse	130-127-575
Sample carriers ⁴	
MACSwell One Imaging Frames	see website
MACSwell One Small Imaging Frames	
MACSwell Two Imaging Frames	
MACSwell Four Imaging Frames	
MACSwell 24 Imaging Plates	
Buffers and accessories	
MACSima Buffer Starting Kit	130-125-753
MACSima Running Buffer 6×1.5 L	130-121-565
MACSima System Buffer 6×1.5 L	130-125-315
MACSQuant®/MACSima Storage Solution 6×1.5 L	130-092-748
MACSwell Deepwell Plates	130-126-865
MACSwell Sealing Foils	130-126-866

¹ The MACSima System is delivered with a trolley table specifically designed to support the system. It has anti-vibration properties and provides a closed storage compartment.

² For detailed information about the different license options, visit [miltenyibiotec.com/MACS-iQ-View](https://www.miltenyibiotec.com/MACS-iQ-View)

³ For details on the wide range of Miltenyi Biotec antibodies for MICS, visit [miltenyibiotec.com/MICS-antibodies](https://www.miltenyibiotec.com/MICS-antibodies)

⁴ For more information about the latest sample carriers from Miltenyi Biotec, visit [miltenyibiotec.com/MICS-sample-carriers](https://www.miltenyibiotec.com/MICS-sample-carriers)



Miltenyi Biotec

Germany/Austria

Miltenyi Biotec B.V. & Co. KG
Friedrich-Ebert-Straße 68
51429 Bergisch Gladbach
Germany
Phone +49 2204 8306-0
Fax +49 2204 85197
macsde@miltenyi.com

USA/Canada

Miltenyi Biotec, Inc.
2303 Lindbergh Street
Auburn, CA 95602, USA
Phone 800 FOR MACS
Phone +1 866 811 4466
Fax +1 877 591 1060
macsus@miltenyi.com

Australia

Miltenyi Biotec
Australia Pty. Ltd.
Unit 11, 2 Eden Park Drive
Macquarie Park, NSW 2113
Australia
Phone +61 2 8877 7400
Fax +61 2 9889 5044
macsau@miltenyi.com

Benelux

Miltenyi Biotec B.V.
Dellaertweg 9C
2316 WZ Leiden
The Netherlands
macsnl@miltenyi.com
**Customer service for:
The Netherlands**
Phone 0800 4020120
Fax 0800 4020100

Belgium

Phone 0800 94016
Fax 0800 99626

Luxembourg

Phone 800 24971
Fax 800 24984

China

Miltenyi Biotec Technology &
Trading (Shanghai) Co., Ltd.
Room A401, 4/F
No. 1077, Zhongheng Road
Pudong New Area
201203 Shanghai, P.R. China
Phone +86 21 6235 1005-0
Fax +86 21 6235 0953
macscn@miltenyi.com.cn

France

Miltenyi Biotec SAS
10 rue Mercœur
75011 Paris, France
Phone +33 1 56 98 16 16
macsfr@miltenyi.com

Hong Kong

Miltenyi Biotec Hong Kong Ltd.
Unit 301, Lakeside 1
No. 8 Science Park West Avenue
Hong Kong Science Park
Pak Shek Kok, New Territories
Hong Kong
Phone +852 3751 6698
Fax +852 3619 5772
macshk@miltenyi.com.hk

Italy

Miltenyi Biotec S.r.l.
Via Paolo Nanni Costa, 30
40133 Bologna
Italy
Phone +39 051 6 460 411
Fax +39 051 6 460 499
macsit@miltenyi.com

Japan

Miltenyi Biotec K.K.
NEX-Eitai Building 5F
16-10 Fuyuki, Koto-ku
Tokyo 135-0041, Japan
Phone +81 3 5646 8910
Fax +81 3 5646 8911
macsjp@miltenyi.com

Nordics and Baltics

Miltenyi Biotec Norden AB
Medicon Village
Scheeleorget 1
223 81 Lund
Sweden
macsse@miltenyi.com
**Customer service for:
Sweden**
Phone 0200 111 800
Fax +46 280 72 99

Denmark

Phone 80 20 30 10
Fax +46 46 280 72 99

**Norway, Finland, Iceland,
and Baltic countries**

Phone +46 46 280 72 80
Fax +46 46 280 72 99

Singapore

Miltenyi Biotec Asia Pacific Pte Ltd.
438B Alexandra Road, Block B
Alexandra Technopark
#06-01
Singapore 119968
Phone +65 6238 8183
Fax +65 6238 0302
macssg@miltenyi.com

South Korea

Miltenyi Biotec Korea Co., Ltd.
Donggeuk 7F,
562 Nonhyeon-ro
Gangnam-gu
Seoul 06136, South Korea
Phone +82 2 555 1988
Fax +82 2 555 8890
macskr@miltenyi.com

Spain

Miltenyi Biotec S.L.
C/Luis Buñuel 2
Ciudad de la Imagen
28223 Pozuelo de Alarcón (Madrid)
Spain
Phone +34 91 512 12 90
Fax +34 91 512 12 91
macses@miltenyi.com

Switzerland

Miltenyi Biotec Swiss AG
Gibelinstrasse 27
4500 Solothurn
Switzerland
Phone +41 32 623 08 47
Fax +49 2204 85197
macsch@miltenyi.com

United Kingdom

Miltenyi Biotec Ltd.
Almac House, Church Lane
Bisley, Surrey GU24 9DR, UK
Phone +44 1483 799 800
Fax +44 1483 799 811
macsuk@miltenyi.com

www.miltenyibiotec.com

Miltenyi Biotec provides products and services worldwide. Visit www.miltenyibiotec.com/local to find your nearest Miltenyi Biotec contact.

Unless otherwise specifically indicated, Miltenyi Biotec products and services are for research use only and not for therapeutic or diagnostic use. MACS, MACSima, MACSQuant, MACSwell, the Miltenyi Biotec logo, REA dye, REAlease, REAfinity, REAlease, and REAscreen are registered trademarks or trademarks of Miltenyi Biotec B.V. & Co. KG and/or its affiliates in various countries worldwide. Copyright © 2023 Miltenyi Biotec and/or its affiliates. All rights reserved.